Highlights

- Four-probe shortcut audit reproduces 99.77% accuracy without the spectrum.
- OOD ladder drops XGBoost from 94% to 56.8% accuracy under LORO.
- Three attribution methods converge on 121–422 Hz as the causal band.
- Audit validated on 9 external benchmarks across 3 sensor modalities.
- Tree ensemble at OOD macro-F1 0.4326, 6.8 ms/1k rows, zero GPU.